Population Parameters and Sample Statistics

Global Economics | Grade 11 | Term 2 | Lesson 1

C1. Distinguishes population parameters from sample statistics by interpreting and comparing the notation for population and sample means, standard deviations, and proportions in economic and financial contexts.

Worked example

1. A market association's sales survey

A market association wants to describe all 1,200 stalls licensed for June. In a random sample of 60 stalls, mean June sales revenue was 4.8 million COP and the sample standard deviation was 1.2 million COP. Eighteen of the sampled stalls accept digital payment.

Task 1. Identify the population and sample. Use μ, s, $\hat{p}$, σ, $\bar{x}$, and p to pair the population and sample measures for mean sales, the standard deviation of sales, and the share of stalls that accept digital payment. Interpret each measure in context and calculate the sample proportion.

Task 2. The association draws another random sample from the same June list. The new sample has mean June sales revenue of 5.1 million COP, a standard deviation of 1.0 million COP, and a digital-payment share of 25%. What changes, what stays the same, and which differences can you calculate? Subtract the first sample's statistics from the second's. Use million COP for sales measures and percentage points for the share. Explain what the two draws establish about the population parameters.

Worked reasoning

Matching the measures. The population is all 1,200 stalls licensed for June. The first sample is the 60 randomly selected stalls. Each row pairs measures of the same feature.

Population parameter	Sample statistic
μ, mean June sales revenue across all 1,200 stalls	$\bar{x} = 4.8$ million COP, mean June sales revenue across the 60 selected stalls
σ, standard deviation of June sales revenue across all 1,200 stalls	$s = 1.2$ million COP, standard deviation of June sales revenue across the 60 selected stalls
p, share of all 1,200 stalls that accept digital payment	$\hat{p} = 18/60 = 0.30 = 30\%$, share of the 60 selected stalls that accept digital payment

Comparing the samples. The draw is new, and its size is unreported. The same June population and measure definitions apply, so μ, σ, and p stay fixed. Their values are unknown from the given summaries.

Second sample minus first	Interpretation
$\bar{x}$: $5.1 - 4.8 = +0.3$ million COP	Higher sample mean sales revenue
s: $1.0 - 1.2 = -0.2$ million COP	Lower sample standard deviation of sales revenue
$\hat{p}$: $25\% - 30\% = -5$ percentage points	Lower sample digital-payment share

Guided practice

2. A supermarket's checkout survey

A supermarket manager wants to describe all 1,500 purchases made at one store on Saturday. In random sample A of 50 receipts, the mean purchase value was 64 thousand COP and the sample standard deviation was 18 thousand COP. Twenty of those purchases used a loyalty card.

Task 1. Identify the population and sample A. Pair the population and sample symbols for mean purchase value, the standard deviation of purchase values, and the loyalty-card share. Explain what each symbol describes and calculate the sample proportion as a percentage.

Task 2. A colleague draws random sample B of 50 receipts from the same store's Saturday purchases. The mean purchase value is 61 thousand COP and the sample standard deviation is 20 thousand COP. Twenty-five of those purchases used a loyalty card. Use the table to compare the two samples. What changes, what stays the same, and which differences can you calculate? Give the difference in loyalty-card shares in percentage points. Explain what the two draws establish about the population parameters.

Measure	Sample A	Sample B	B minus A
Mean (thousand COP)	64	61	
Standard deviation (thousand COP)	18	20	
Loyalty-card share			

Independent practice

Complete your answers on a separate sheet.

3. A coffee association's export shipments

A coffee exporters' association wants to describe all 900 coffee shipments in its April export records. In a random sample of 40 shipments, the mean weight was 18.4 metric tonnes and the sample standard deviation was 4.0 metric tonnes. Twelve of the sampled shipments departed on schedule.

Task 1. Identify the population and sample. Pair the population and sample symbols for the mean, standard deviation, and proportion. Interpret each measure in context and calculate the sample proportion.

Task 2. A second analyst draws a fresh random sample of 40 shipments from the same April records. Its mean shipment weight is 19.1 metric tonnes and its sample standard deviation is 3.6 metric tonnes. Twelve of these shipments departed on schedule. What changes, what stays the same, and which differences can you quantify? Use appropriate units, including percentage points for differences in proportions. Separate changes in sample statistics from what the data establish about the population parameters.

4. A transit agency's fare review

A transit agency wants to describe all 8,400 bus boardings on its central routes last Monday. In a random sample of 100 boarding records, the mean fare paid was 3.2 thousand COP and the sample standard deviation was 0.8 thousand COP. Thirty-five of the sampled boardings used a discounted fare.

Task 1. Identify the population and sample. Pair the population and sample symbols for the mean, standard deviation, and proportion. Interpret each measure in context and calculate the sample proportion.

Task 2. The director adds suburban routes to the review for the same Monday. The combined routes had 12,000

boardings. A random sample of 100 from the combined records gives a mean fare of 3.4 thousand COP and a sample standard deviation of 0.9 thousand COP. Thirty sampled boardings used a discounted fare. What changes, what stays the same, and which differences can you quantify? Use appropriate units, including percentage points for differences in proportions. Separate changes in sample statistics from what the data establish about the population parameters.

5. A credit union's loan review

A credit union wants to describe all 800 of its active microenterprise working-capital loans on June 30. In a random sample of 50 loans, the mean outstanding balance was 8.0 million COP and the sample standard deviation was 2.4 million COP. Ten of those loans had an overdue installment on June 30.

Task 1. Identify the population and sample. Pair the population and sample symbols for the mean, standard deviation, and proportion. Interpret each measure in context and calculate the sample proportion.

Task 2. The lending committee adds active microenterprise equipment loans to the June 30 review, bringing the total to 1,000 loans. A random sample of 50 from the combined list has a mean outstanding balance of 8.8 million COP and a sample standard deviation of 2.9 million COP. Eight of those loans had an overdue installment on June 30. What changes, what stays the same, and which differences can you quantify? Use appropriate units, including percentage points for differences in proportions. Separate changes in sample statistics from what the data establish about the population parameters.

Solutions

Sample (green)	Population (red)
Sample mean ($\bar{x}$)	Population mean (μ)
Sample standard deviation (s)	Population standard deviation (σ)
Sample proportion ($\hat{p}$)	Population proportion (p)
Sample (selected units)	Population (all units of interest)

2. A supermarket's checkout survey

Matching the measures. The population is all 1,500 Saturday purchases at the store. Sample A is the 50 purchases represented by the selected receipts. Each row pairs measures of the same feature.

Population parameter	Sample statistic
μ, mean purchase value across all 1,500 purchases	$\bar{x}$ = 64 thousand COP, mean purchase value across the 50 sampled purchases
σ, standard deviation of purchase values across all 1,500 purchases	s = 18 thousand COP, standard deviation of purchase values across the 50 sampled purchases
p, share of all 1,500 purchases that used a loyalty card	$\hat{p} = 20/50 = 40\%$, share of the 50 sampled purchases that used a loyalty card

Comparing the samples. B is a new random draw of 50 receipts from the same Saturday population. The same measures apply, so μ, σ, and p stay fixed. Their values are unknown from the given summaries. The loyalty-card count increases by $25 - 20 = 5$ purchases, and B's loyalty-card share is $25/50 = 50\%$.

Sample B minus Sample A	Interpretation
$\bar{x}$: $61 - 64 = -3$ thousand COP	Lower sample mean purchase value
s: $20 - 18 = +2$ thousand COP	Higher sample standard deviation
$\hat{p}$: $50\% - 40\% = +10$ percentage points	Higher sample loyalty-card share

3. A coffee association's export shipments

Task 1. The population is all 900 April shipments in the association's records. The sample is the 40 selected shipments. For all 900 shipments, μ is the mean shipment weight, σ is the standard deviation of shipment weights, and p is the share that departed on schedule. For the 40 sampled shipments, the corresponding mean is $\bar{x}$ = 18.4 metric tonnes, the standard deviation is s = 4.0 metric tonnes, and the on-schedule share is $\hat{p} = 12/40 = 30\%$.

Task 2. The draw is new. The population, sample size, and on-schedule count stay the same. The second sample's mean is higher by $19.1 - 18.4 = 0.7$ metric tonnes, and its standard deviation is lower by $4.0 - 3.6 = 0.4$ metric tonnes. Its on-schedule share is $12/40 = 30\%$, a difference of 0 percentage points. The three population parameters stay fixed because both draws refer to the same 900 shipments. Their values remain unknown from the summaries.

Solutions (continued)

4. A transit agency's fare review

Task 1. The population is all 8,400 Monday boardings on the central routes. The sample is the 100 boardings represented by the selected records. Across the 8,400 boardings, μ is the mean fare paid, σ is the standard deviation of fares paid, and p is the share that used a discounted fare. Among the 100 sampled boardings, the corresponding mean is $\bar{x} = 3.2$ thousand COP, the standard deviation is $s = 0.8$ thousand COP, and the discounted-fare share is $\hat{p} = 35/100 = 35\%$.

Task 2. The population now covers central and suburban routes on the same Monday. It increases by $12{,}000 - 8{,}400 = 3{,}600$ boardings. The sample is a new draw of the same size, 100 boardings, from the combined records. Its discounted-fare count falls by $35 - 30 = 5$ boardings.

The sample mean fare rises by $3.4 - 3.2 = 0.2$ thousand COP and the sample standard deviation rises by $0.9 - 0.8 = 0.1$ thousand COP. The new sample share is $30/100 = 30\%$, so the difference is $30\% - 35\% = -5$ percentage points.

The parameter symbols keep the same meanings for fare and discount use, with the combined boardings as their new population. The sample summaries leave open whether the population parameters differ and, if so, by how much.

5. A credit union's loan review

Task 1. The population is all 800 active microenterprise working-capital loans on June 30. The sample is the 50 selected loans. Across all 800 loans, μ is the mean outstanding balance, σ is the standard deviation of outstanding balances, and p is the share with an overdue installment on June 30. Among the 50 sampled loans, the corresponding mean is $\bar{x} = 8.0$ million COP, the standard deviation is $s = 2.4$ million COP, and the overdue share is $\hat{p} = 10/50 = 20\%$.

Task 2. The population now covers both active working-capital and equipment loans on June 30. It increases by $1{,}000 - 800 = 200$ loans. The sample is a new draw of the same size, 50 loans, from the combined list. Its overdue count falls by $10 - 8 = 2$ loans.

The sample mean balance rises by $8.8 - 8.0 = 0.8$ million COP and the sample standard deviation rises by $2.9 - 2.4 = 0.5$ million COP. The new sample share is $8/50 = 16\%$, so the difference is $16\% - 20\% = -4$ percentage points.

The parameter symbols keep the same meanings for balances and overdue installments, with both loan categories as their new population. The sample summaries leave open whether the population parameters differ and, if so, by how much.